KULLIYYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY
FINAL YEAR PROJECT 1 (FYP1)
PROJECT PROPOSAL

To be prepared BEFORE approaching a potential supervisor

Course Code	☒ BICS 4321 ☐ CSCI 4401
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Proposed Project Title	BagSense
Project Type	☒ System Development Project ☐ Research Project
Project Category / Area	Internet of Things (IoT) / Embedded Systems

IMPORTANT

Maximum length: 6 pages. Complete all relevant sections before contacting a potential supervisor. Do not list a supervisor's name until the staff member has agreed to supervise the project.

1. Project Overview

Project BagSense is an energy-efficient Edge-IoT smart backpack that performs autonomous, non-line-of-sight inventory tracking inside the bag cavity. It integrates a near-field Ultra-High Frequency (UHF) passive RFID reader with Hall-effect zipper-state sensing so that tagged personal items can be inventoried without optical line of sight and without unpacking the bag. User interaction is provided through a chassis-mounted low-power E-paper display for glanceable status and a companion cross-platform Flutter application connected over Bluetooth Low Energy (BLE) and Wi-Fi. The system is targeted at university students and mobile professionals, and its application area is inventory containment and item accountability at the point of packing.

2. Background of the Study / Project

Students at IIUM move daily between lecture halls, laboratories, cafes and residential colleges across the KICT complex. That constant transition makes it easy to leave a matric card, charger or textbook behind, and the loss is only discovered after the student has already arrived at the next venue.

Existing consumer trackers such as the Apple AirTag and Samsung SmartTag solve single-item recovery, but each tagged item needs its own battery and between RM120 and RM160 of hardware, which makes tagging a full set of personal belongings economically impractical. This leaves a clear operational gap: there is no ambient, low-power smart container that automates item accountability at the moment of packing.

3. Problem Statement / Problem Description

Existing approaches to personal item tracking suffer from three structural weaknesses:

1. High per-item cost and battery dependence. Active Bluetooth beacon tags such as AirTag and SmartTag cost RM120–RM160 per unit and rely on internal coin cells. Tagging ten common items in one backpack therefore demands over RM1,200 in consumable tags alone.

2. Optical occlusion in closed bags. Camera-based approaches placed inside a bag fail under extreme occlusion, absent ambient illumination and erratic item folding.

3. Passive, friction-heavy verification. Verification of what is in the bag requires manual searching or unpacking, and users typically discover a missing item only after arriving at a lecture or examination venue.

4. Project Objectives

The technical objective is to design, build and validate a low-cost, production-feasible smart backpack that automates item awareness. The specific measurable objectives are:

1. Multi-Tag Non-Line-of-Sight Inventory Engine — to implement and optimise anti-collision UHF RFID interrogation that inventories up to 15 tagged items inside the enclosed bag cavity in under 1.5 seconds, with recall of at least 98% (both figures to be confirmed by the pilot measurement in Section 9 before they are treated as design commitments).

2. Dual-Interface Visualisation — to develop a low-power external E-paper display for glanceable status outside the bag, together with a cross-platform Flutter application for checklist creation, missing-item alerts and event logs.

3. Power-Aware Edge Firmware — to design an event-driven ESP32-S3 firmware architecture using deep-sleep states and Hall-effect zipper switches, targeting more than 5 days of operation per charge.

4. Read-Zone Confinement and Field Validation — to design the cavity liner and antenna placement so that the interrogating field stays inside the bag, and to benchmark recognition accuracy, read-zone confinement and usability in the KICT building.

5. Research Questions (Research Project Only)

Not Applicable

6. Scope of the Study / Project

1. Target Users. University students, lecturers and academic staff who routinely carry multiple study and technological items between campus venues.

2. Main Functions.

- In-cavity inventory scan: up to 15 passive RFID-tagged items identified without optical line of sight in under 1.5 seconds.
- Event-driven sensing: Hall-effect zipper switches detect opening and closing and trigger the scan routine.
- Dual-channel status output: glanceable present/missing indicators on an external E-paper panel, with itemised checklist logs synchronised to the smartphone application.
- Power management: deep-sleep between zipper events to extend battery operation.

3. Platform and Hardware Specifications.

- Embedded edge controller: ESP32-S3 (dual-core 240 MHz, BLE 5.0 + Wi-Fi) running FreeRTOS tasks in C++.
- Scanning hardware: YRM100 / MagicRF Nano UHF RFID reader module configured for the Malaysian 919–923 MHz band (MCMC SRSP 530), with a near-field UHF loop antenna tuned to the same band, plus a 2.9-inch SPI E-paper panel.
- Mobile application: Flutter (Dart) front-end for Android and iOS.
- Mechanical chassis: standard multi-compartment student backpack retrofitted with 3D-printed snap-fit enclosures modelled in Blender.

4. Data Source. Data is gathered from passive UHF RFID tag responses (Impinj Monza / Alien Higgs-3 transponders) inside the bag cavity and from Hall-effect state transitions. Packing lists and user preferences are stored locally in the Flutter application.

5. Deployment Context. Testing and deployment are bounded to campus environments, specifically lecture halls, study areas and laboratories within the KICT building.

6. Project Boundaries.

- No global GPS tracking: the system addresses inventory containment inside the bag and does not incorporate real-time outdoor GPS or cellular positioning.
- No active per-item tracking: the project does not use active BLE or UWB tags such as AirTag; tracking is limited to passive, batteryless UHF RFID labels.
- No reading of tags outside the bag: the read zone is deliberately confined to the cavity by a near-field loop antenna mounted inside the cavity and a shielding liner applied to the outer shell (body side and outside panels). The inner cavity lining remains RF-transparent so that the field reaches every tagged item, and the BLE antenna is mounted on the unshielded top flap so that the smartphone link is not attenuated.

7. Significance / Expected Contribution

1. Cost efficiency. Item-level tracking is lowered to passive RFID labels costing under RM1.00 each, against RM120+ for active beacons, which makes tagging an entire bag economically reasonable.

2. Frictionless usability. Item verification is automated through zipper open/close events, removing the need to unpack or to open the smartphone, since status is visible on the external E-paper panel.

3. Academic and engineering contribution. The project demonstrates a practical edge implementation of near-field UHF RFID inside a wearable, flexible container, and documents the electromagnetic confinement and low-power battery management trade-offs involved.

8. Brief Review of Related Work

1. Smart luggage tracking using IoT and GPS (Jain et al., 2023). A microcontroller, GPS and GSM luggage system for tracing missing baggage. Effective for locating a bag as a whole, but it monitors no contents and cannot detect an individual missing item inside the compartment.

2. Energy-aware task scheduling for batteryless IoT (Sabovic et al., 2022). Scheduling of application tasks under intermittent harvested energy with defined voltage thresholds. This supplies the rationale for BagSense's event-driven architecture, which pairs deep sleep with magnetic zipper triggers to limit active consumption.

3. Near-field UHF RFID antennas in wearable and flexible contexts (Daiki et al., 2014; Lalkota et al., 2018). Both works show that UHF reader antennas can be shaped to operate in the near field, and that near-field and far-field behaviour are distinct design targets. This supports the choice of a loop-type antenna confined to the cavity rather than a conventional radiating patch.

4. RFID inventory accuracy in confined read zones (Hellström & Wiberg, 2010). A case study showing that read accuracy in practice is governed by tag placement, material loading and reader configuration rather than by the reader's nominal range. This is the basis for treating the 98% recall figure as a target to be measured, not assumed.

Gap. The reviewed work either locates a bag as a whole, or reads tags on shelves and pallets. None addresses inventory of loose, folding personal items inside a small flexible cavity that the user carries against their body, where the field must be confined to avoid reading bystanders' tags while still reaching every item. That is the gap BagSense addresses.

9. Proposed Methodology / Development Approach

The project follows an Agile prototyping methodology across four phases, with a pilot measurement placed before the evaluation phase so that the performance targets are set from measured data.

Phase 1 — Requirements and hardware dimensioning. Measure the internal dimensions of representative student backpacks, map candidate tag positions, finalise the power budget and select parts against the 919–923 MHz band.

Phase 2 — Enclosure and liner design (CAD). Model snap-fit electronics housings, cable conduits, the external display bezel and the shielding liner placement in Blender, then 3D-print and test-fit them.

Phase 3 — Firmware and software implementation. Develop FreeRTOS tasks on the ESP32-S3 for reader communication, Hall-effect interrupt handling, E-paper rendering and a BLE GATT server; build the Flutter application for checklist configuration and event logs.

Phase 4 — Pilot measurement, integration and evaluation. First run a pilot with one antenna, five tagged items and three positions over repeated trials to establish the achievable recall and latency envelope. Then run the full benchmark: accuracy against tag count and position, field confinement tested with tags outside the bag at 0.5 m and 1.0 m, battery endurance over a five-day campus routine, and a usability evaluation with ten participants using the System Usability Scale.

Division of work. Student 1 (2418711) leads firmware, reader integration, antenna and read-zone characterisation, and the evaluation. Student 2 (2419467) leads enclosure and liner CAD, 3D printing, the Flutter application and field data collection. Both students share integration testing and reporting.

Timeline. Semester 1 covers Phases 1 to 3 and the pilot measurement; Semester 2 covers the evaluation phase, refinement and the final report.

10. Expected Output / Deliverables

1. BagSense prototype. A retrofitted smart backpack integrating the ESP32-S3 controller, near-field UHF RFID module and loop antenna, Hall-effect zipper sensors and power management circuit.

2. Display and bezel assembly. A 3D-printed outer bezel mounting the 2.9-inch SPI E-paper status panel.

3. Companion Flutter application. An Android/iOS application supporting custom packing lists, category tags and BLE synchronisation.

4. Firmware and evaluation report. The complete ESP32-S3 firmware codebase together with an evaluation report reporting the measured accuracy, latency, read-zone confinement and battery endurance, assessed against the targets of ≥ 98% recall, ≤ 1.5 s and ≥ 5 days.

REFERENCES

Daiki, M., Perret, E., & Tedjini, S. (2014). Design of near field UHF RFID reader antenna integrated into clothing. In 2014 IEEE RFID Technology and Applications Conference (RFID-TA) (pp. 261–265). https://doi.org/10.1109/RFID-TA.2014.6934240

Hellström, D., & Wiberg, M. (2010). Improving inventory accuracy using RFID technology: A case study. Assembly Automation, 30(4), 345–351. https://doi.org/10.1108/01445151011075807

Jain, S., Aithal, S., Allamsetty, V., Sarvikasree, K. S., Gobi, N., & Rathi, R. (2023). Smart luggage tracking using IoT and GPS technology. International Journal of Advanced Research in Computer and Communication Engineering, 12(4). https://doi.org/10.17148/IJARCCE.2023.12403

Lalkota, M., Gupta, G., & Harish, A. (2018). UHF reader antenna for near and far field RFID operation. In 2018 IEEE International Conference on RFID (RFID) (pp. 1–7). https://doi.org/10.1109/RFID.2018.8376196

Sabovic, A., Sultania, A. K., Delgado, C., De Roeck, L., & Famaey, J. (2022). An energy-aware task scheduler for energy-harvesting batteryless IoT devices. IEEE Internet of Things Journal, 9(22), 23097–23114. https://doi.org/10.1109/JIOT.2022.3185321

Student Readiness Checklist

☒ I have earned at least 75 credit hours ☒ I have passed the prerequisite courses: BICS 2302 Data Structures and Algorithms; BICS 2303 Intelligence Systems

Note: This proposal is intended for initial supervisor consideration and may be refined after discussion with the appointed supervisor.